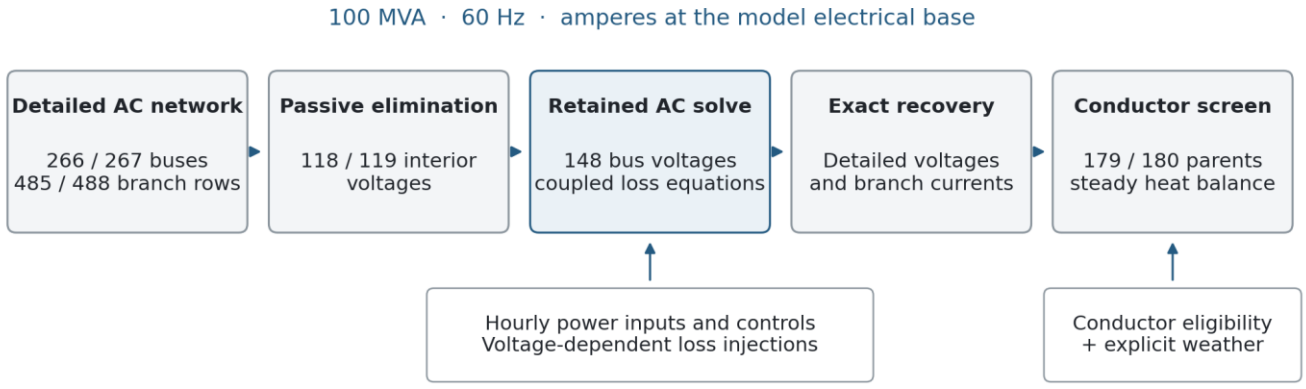
NYgrid 2025 corrected model structure

Electrical network and conductor thermal representation

This document describes the model supplied for continued dynamic line rating research: a 148-bus, loss-consistent AC power-flow model with recoverable detailed branch currents and conductor realizations constrained by electrical resistance and reactance. It explains the current network, operating inputs, equations, controls and thermal interpretation. Package setup and run commands are in START_HERE.md.

The compact and detailed networks are two representations of the same current model. Four dated topology states select the network applicable to each 2025 hour. The case is a synthetic research equivalent; its electrical and thermal assumptions do not establish an observed NYISO operating state or installed asset rating.

Quantity	Current structure
Retained AC buses	148 in every dated state
Detailed AC buses	266 before full-service Dover; 267 afterward
Detailed AC branch rows	485 before full-service Dover; 488 afterward
Parent corridor records	410 before full-service Dover; 413 afterward
Generator rows	1,464, with hourly status and bounds
Prescribed transfer rows	6 MATPOWER DC-line records
Time axis	8,760 hours; 8,651 source-eligible hours
Electrical base	100 MVA; bus-specific nominal kV; 60 Hz



Parent corridors identify modeled facilities or equivalents. Detailed branch rows include numerical line sections and equipment records. Reduced admittance couplings are mathematical objects and have no independent conductor or thermal rating.

Network and dated topology

The 148 retained buses comprise 54 New York, 36 ISONE, 31 Ontario and 27 PJM buses. They include 140 NPCC boundary and internal buses plus eight New York 345-kV station ports. The additional ports retain junction voltages without assigning them extra demand or generation solely because they are retained.

Retained station port	Bus ID	Retained station port	Bus ID
East Fishkill	10858	Wood Street	10902
Ladentown	10774	Dunwoodie	10593
East Garden City	10589	Gowanus	11129
Rainey	11130	Cricket Valley	20001

In the detailed representation, 43 additional Downstate terminals and 75 numerical section buses are passive and eliminated. The post-Dover state adds one passive regulated port. Thus 118 or 119 eliminated voltages supplement the same 148 retained voltages. Numerical section buses represent discretization, not extra substations.

State begins	Topology state	Detailed buses / branches
Start of local 2025	pre_may	266 / 485
21 May 2025 00 EDT	may21	266 / 485
30 May 2025 00 EDT	may30	266 / 485
19 Dec 2025 23 EST	post_dover	267 / 488

The May boundaries activate the modeled Gowanus–Greenwood and Goethals–Fox Hills receiving paths. The Dover boundary is 20 December 2025 at 04 UTC. These dates are implemented switching policies. An hour selects exactly one state; the states are not separate model versions.

Detailed branch representation

The detailed network contains overhead corridors, cable and urban equivalents, transformer-compatible paths, composite voltage-changing equivalents, PAR equipment and inactive records. The section map records each parent, its numerical sections and its physical from/to terminals. Inactive rows remain in the axis and must be distinguished from active zero-flow rows.

Fixed shunts and declared charging compensation remain part of the admittance model. Parent 224 is a released Gilboa–Buchanan network equivalent with total R = 0.02 pu, X = 0.0147 pu and B = 0; it remains electrical-only under the current conductor catalog and geometry limits.

Declared physical parameters

Parent 41 uses imported R = 0.0018 pu and X = 0.0089 pu with a coherent synthetic conductor realization. Its charging B is derived from the same geometry. Candidate selection uses the predeclared imported impedance inference and original charging only as a ranking aid. Neither observed interface errors nor a map chord selects these parameters.

Parent 41 quantity	Current value
Conductor family and catalog	ACSS Mallard 795 kcmil
Parallel circuits and bundle	4 circuits and 1 subconductor per phase
Equivalent length	10.750873 km
Phase GMD	3.948477 m
Reference and maximum temperature	95 degrees C
Total R and X	0.0018 and 0.0089 pu
Total derived charging B	0.02126747227 pu

This is an admissible equivalent realization, not an identification of installed conductors or physical route length. The aggregate apparatus current ceiling remains unchanged. The model registry and thermal segment records carry the current parameter basis and broad synthetic length bounds.

Section local compensation

Parents 185 and 195 retain their conductor parameters and their total fixed reactor rating of 75 percent of geometric line charging. Each pi section places half of its local 75 percent compensation at each end; contributions add at shared internal nodes. This distributed placement is a declared synthetic design. It does not identify installed intermediate stations or adjustable controllers.

Parent	Sections	Total reactor at 1 pu MVAr	Each terminal MVAr
185	6	607.279468	50.606622
195	7	254.817180	18.201227

The full-axis bus shunt vector and reactor ledger preserve this placement. Internal section buses remain passive nodes with zero prescribed injection, so their fixed shunts stay in the exact admittance reduction. Voltage limits, PV setpoints, hourly generation, demand and transfer targets are retained. The corrected admittance and current operators are rebuilt from the resulting case.

Generation and controlled transfers

The 1,464 generator rows contain 1,436 native resource rows, 27 external regional equivalents and one signed HQ equivalent. A generator row is distinct from its electrical bus ID and reporting identity. Multiple rows can share a bus. Hourly active-power schedules, capability bounds and status are supplied on a fixed row axis; the active count changes with the hour.

Active and reactive controls

Native real-power schedules and the signed HQ injection are fixed in the loss-consistent solve. ISONE, Ontario and PJM equivalents provide their assigned AC losses using fixed participation weights and finite original real-power bounds. External delivered-load proxies remain fixed. The reference angle is set at bus 50; a numerical reference source is not an unlimited provider of active-power losses.

PV buses retain their stated voltage magnitudes. The solver determines the associated reactive injections without enforcing generator Q limits through bus-type switching. The inherited capability assumptions use a 0.95 power-factor envelope and apparent-power circles where specified. The audit separately checks the 0.95–1.05 pu voltage range, reactive bounds and generator capability.

Prescribed exchanges

Six MATPOWER DC-line records prescribe active-power transfers: four external-transfer representations and two Ontario PAR-flow substitutions. They are six modeling records, not six asserted physical HVDC assets. Their scheduled signs, endpoint injections and reactive conventions are retained. Their declared real-power losses are zero in the current accounting policy.

Channel	Model from bus	Model to bus
CSC	21 MILLSTONE PT	80 NORTHPORT
NEPTUNE	124 BRANCHBURG	79 RAV A-3
HTP and VFT	125 LINDON	81 GOETHALS
Ontario PAR 25118 and 25119	100 PHAS	48 MOSES E

These are model endpoint buses, not asserted literal asset terminals. Positive transfer is from-to; a negative target reverses it. Endpoint reactive injections are fixed at zero for these six records.

The Ontario substitutions disable the replaced aggregate AC transfer path. In the post-Dover state, two separate parallel PAR-bank records use the bounded hourly phase reconstruction. These prescribed controls condition the power flow; the solve does not optimize their schedules or infer actual tap positions.

Meaning of the input dispatch

The native dispatch is source-conditioned and capacity bounded. It is not a unit-commitment solution: ramps, reserves, startup decisions, minimum stable output and storage energy trajectories are not established. External demand and reactive behavior are synthetic assumptions. The model is suitable for controlled research scenarios only when these assumptions are retained or explicitly replaced.

AC equations and exact recovery

Let V be the complex bus-voltage vector, Y the detailed bus-admittance matrix and S the net complex injection on the stated base. The coupled solver evaluates voltage-dependent delivered demand and external loss participation together with the AC equations.

$$S(V) = diag(V)\,(YV)^*$$

Partition buses into retained r and eliminated e. The eliminated buses have zero independent complex injection under the frozen placement. Their charging and fixed shunts remain in Y. Eliminating those voltages gives a complex-admittance reduction rather than a DC susceptance approximation.

$$V_e = -{Y_{ee}}^{-1} Y_{er} V_r$$

$$Y_{red} = Y_{rr} - Y_{re} {Y_{ee}}^{-1} Y_{er}$$

The 148 retained voltages include a fixed reference angle and controlled PV magnitudes; they are not 296 free real unknowns. Bus types depend on hourly generator admission. A linear lift L reconstructs all detailed voltages. The current operators include branch series admittance, charging, transformer taps and phase shifts, preserving full-network bus and branch ordering.

$$V = LV_r; \quad I_f = Y_f L V_r; \quad I_t = Y_t L V_r$$

A separate series-current operator removes the terminal charging contribution for conductor Joule heating. Terminal currents remain relevant to terminal equipment and the declared rating screen. Current conversion uses the nominal voltage at the appropriate endpoint and the three-phase electrical base.

$$I_{base} = \frac{100 \times 10^6}{\sqrt{3} \times V_{base,kV} \times 10^3}$$

All variable loss-dependent injections remain on retained buses. Adding a new load, generator or controller to an eliminated bus changes the reduction contract and requires rebuilding the retained set and operators. Thermal calculations use recovered detailed currents, never reduced admittance edges.

Hourly inputs and loss accounting

The model stores one annual axis of 8,760 UTC-referenced timestamps spanning local calendar year 2025. A source mask admits 8,651 hours and excludes 109. The bundle contains zonal and bus demand allocations, generator schedules and bounds, prescribed transfers, phase controls, state IDs and eligibility flags. Excluded hours are not replaced with artificial zero inputs.

Power scale and source meaning

The fixed power scale is gamma = 0.30012075175285535, based on a declared 9,561-MW model anchor divided by the 31,857.1773-MW maximum simultaneous public zonal load. Model MW equals gamma times public MW. Aggregate model MW may be divided by gamma for public-series comparison. AC amperes are never divided by gamma.

NY zonal actual load is interpreted as a generation-plus-net-interchange balance. The current model partitions its frozen scaled target into delivered demand and assigned AC losses. Delivered demand is an inferred model quantity, not an observed end-use load series. Within-zone spatial shares and reactive power factors are fixed assumptions.

$$P_{served,z} + P_{loss,z}(V) = P_{gross,z}$$

Within-NY corridor losses are shared equally by endpoint zones; a corridor within one zone assigns its full loss there. External internal corridors and NY-border corridors assign their losses to the appropriate external region. HQ is fixed. These are declared accounting rules, not observed meter allocations.

Simultaneous numerical solve

The solver jointly updates retained voltages and loss-dependent injections using analytic derivatives and a domain-preserving line search. It rejects negative NY delivered demand and external participation that exceeds finite bounds. It does not clip losses, expand capability or fall back to unlimited reference-generator supply.

Successful accounting requires both the AC equations and regional/boundary identities to close. A failed line search or a delivered-load boundary is a failed attempted solve, not proof that every possible operating solution is infeasible. Failed iterates and excluded source hours must not be treated as valid solved states.

Conductor and thermal representation

A synthetic conductor realization is acceptable without identifying an installed circuit. Each accepted overhead realization specifies catalog material, resistance temperature, diameter, GMR, circuit count, bundle count, phase geometry and equivalent length. Its forward resistance and reactance must match the current electrical section. Resistance and reactance are held fixed during selection; charging mismatch is disclosed separately.

$$R = \frac{r_{sub}(T)\,\ell}{n_c n_b}; \quad X = \frac{2\pi f \times 2 \times 10^{-7}\,\ell}{n_c}\, ln(D_m / GMR_{bundle})$$

Here R and X are ohms, length is metres and r_sub is ohms per metre. Convert per-unit impedance using Z_base = V_base,kV² / baseMVA. n_c counts parallel circuits and n_b counts subconductors per phase. X/R constrains geometry; R fixes length for a chosen multiplicity. Several conductor, bundle, circuit and length combinations may reproduce one R/X pair. Catalog temperature and voltage-dependent geometry/length limits define the admissible synthetic family.

The selector also requires active same-voltage overhead sections with compatible taps, phase shifts and consistent segment metadata. Known cable conflicts at parents 300, 309, 310 and 386 are excluded. Parent 224 fails the current 345-kV catalog/geometry X/R condition; this is a conditional no-fit result, not a universal impossibility proof.

Runtime topology	Eligible parents	Branch rows	Material segments
Before full-service Dover	179	251	251
After full-service Dover	180	252	253

Current electrical parameters include both realizations preserving imported R/X and declared engineering constructions. The eligibility ledger records that distinction. The screen does not change either category merely to produce an acceptable thermal utilization.

$$I_{sub}{}^2 r_{sub}(T) + q_{solar} = q_{conv}(T) + q_{rad}(T)$$

Explicit air temperature, wind speed, wind angle, solar irradiance and pressure drive the steady heat balance. Conductor current is the recovered series current divided by n_c n_b. Material segments sharing one lumped series element share that element current; successive electrical pi sections use their own recovered currents. The segment temperature limit determines ampacity. This is a one-way thermal screen, not an R(T)-coupled power flow or DLR optimization.

Data axes and interpretation limits

Object or axis	Meaning
148 retained bus rows	Bus IDs and retained voltages; row number is not the bus ID.
Detailed bus and branch rows	266/267 voltages and 485/488 original-section currents for the selected state.
1,464 generator rows	Fixed generator identity axis with corrected electrical landing and hourly capability.
Parent and section maps	Parent ID, branch key, numerical sections and physical terminal markers.
Thermal segment rows	Catalog, geometry, material, reference temperature and section-current membership.
Hour index and timestamp	One of 8,760 positions, accompanied by source eligibility and topology state.

Separate qualification flags

result.success and audit.loss_allocation_gate establish numerical and accounting closure; audit.reduction_and_equation_gate checks recovered detailed equations. audit.VQ_and_generator_assumption_gate checks operating limits; audit.passed additionally requires those limits. Thermal eligibility establishes a synthetic conductor mapping; utilization evaluates specified weather. None of these flags alone establishes all the others.

The source eligibility mask does not certify an AC operating point. A point may satisfy the coupled equations and loss balances while violating the voltage or generator limits, or while exceeding a scenario ampacity. Preserve each qualification flag alongside its outputs rather than combining them into a single unqualified success label.

The model supports reproducible studies of its AC equations, boundary accounting, reduction and assumed conductor thermal behavior. It does not establish actual critical-span weather, installed asset identity, terminal-limited facility ratings, a feasible dispatch history, N−1 security or economic benefit from DLR. Scenario changes must retain the electrical and thermal bases and must be requalified.

Machine-readable structure

The package includes the current model bundle and its receipt, model/structure ledgers, DATA_DICTIONARY.md and an integrity manifest. The structure ledgers provide the full bus, parent, section, generator and transfer axes behind this description. START_HERE.md contains the public entry points and execution examples; this document is limited to the model itself.

Annual validation and use

The completed fresh annual coupled AC power-flow replay evaluates all 8,651 source-eligible hours on the preserved 8,760-hour axis. The compiled package preserves the hourly input shards byte for byte and uses the same four corrected templates. The public interface definitions, observation series, clock and power scale remain frozen.

Qualification measure	Hours
Full time axis	8760
Source eligible	8651
AC and loss equations	8651
Voltage and generator assumptions	0
Full declared operating gate	0
Solver exceptions	0

These flags answer different questions. Equation and loss closure supports a numerical flow result. Voltage, reactive and generator apparent-power checks test the declared operating assumptions. The full operating gate also includes the remaining modeled equipment and transfer checks. No count establishes installed facility ratings, actual dispatch feasibility, contingencies or an economic benefit from dynamic line rating.

The offline map shows predicted interface flows, delivered zonal load and assigned losses only where the equation and loss mask passes. Frozen gross load, native generation and prescribed transfers use the source mask. Failed iterates and source-excluded hours remain unavailable rather than being interpolated or shown as zero.

The annual comparison uses the same signed original-terminal interface operators and divides aggregate model MW by the fixed gamma for the public comparison. Separate files preserve the paired baseline cohort, new successful hours, lost successes and all-success cohort. Differences across cohorts must not be interpreted as paired model changes.

Exact model and candidate hashes, the annual and scoring receipts, structure ledgers and runtime checks are supplied with the package. The original baseline model and its earlier manual remain in the preserved baseline archive. The current package contains the corrected model described here.